

Visualizations

2026-03-30

Importing Libraries

```
library(ggplot2)
library(dplyr)
library(reshape2)
library(tidyverse)
library(forecast)
```

Importing and Cleaning Data

```
df <- read.csv("Economy_of_US.csv")

df <- df[, c(1, 2, 3, 6, 7, 8, 9)]
colnames(df)[2] <- "GDP"
colnames(df)[3] <- "GDP_PerCapita"
```

Data Visualizations

GDP Year-By-Year, with Significant Events

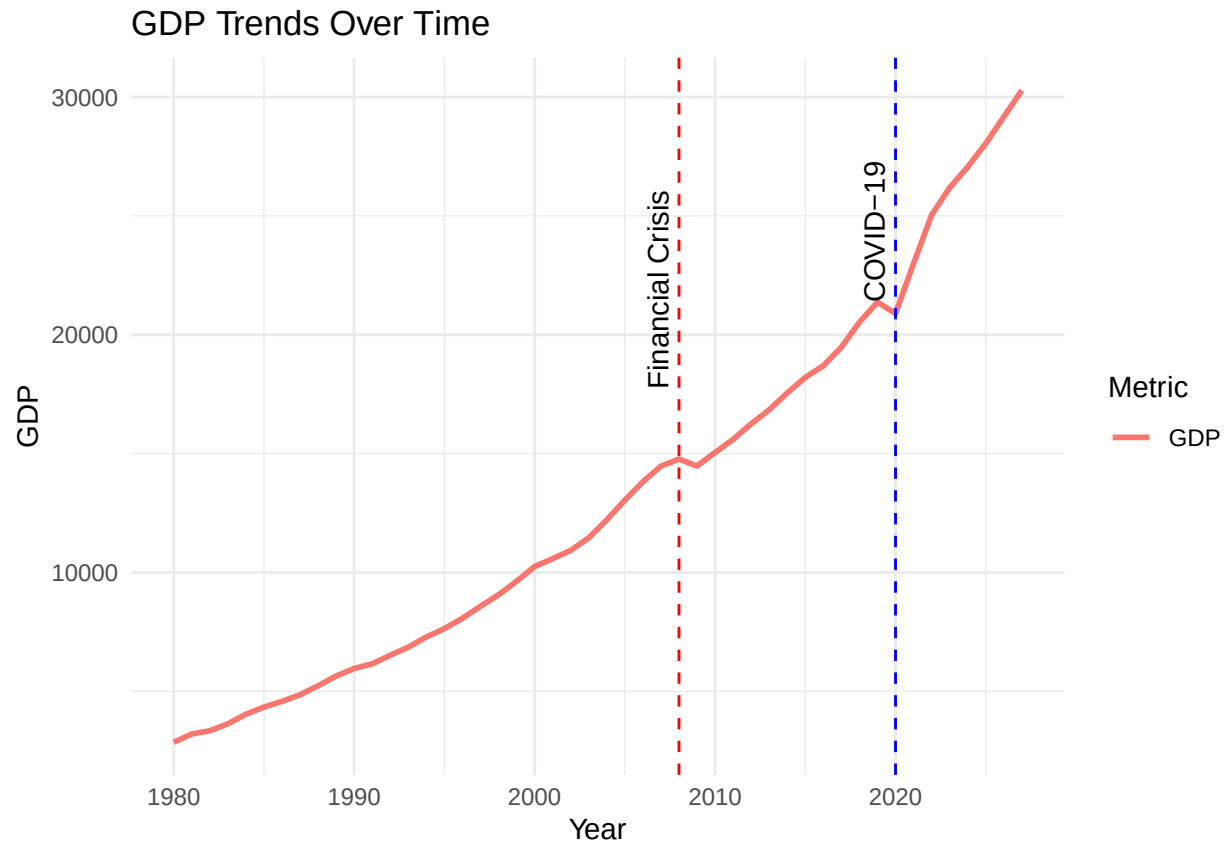
```
df_long <- df %>%
  select(Year, GDP) %>%
  pivot_longer(-Year, names_to = "Metric", values_to = "Value")

ggplot(df_long, aes(x = Year, y = Value, color = Metric)) +
  geom_line(linewidth = 1) +

  # Event annotations
  geom_vline(xintercept = 2008, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 2020, linetype = "dashed", color = "blue") +

  annotate("text", x = 2008, y = max(df$GDP), label = "Financial Crisis", angle = 90, vjust = -0.5, hjust = 1.5) +
  annotate("text", x = 2020, y = max(df$GDP), label = "COVID-19", angle = 90, vjust = -0.5, hjust = 1.5) +

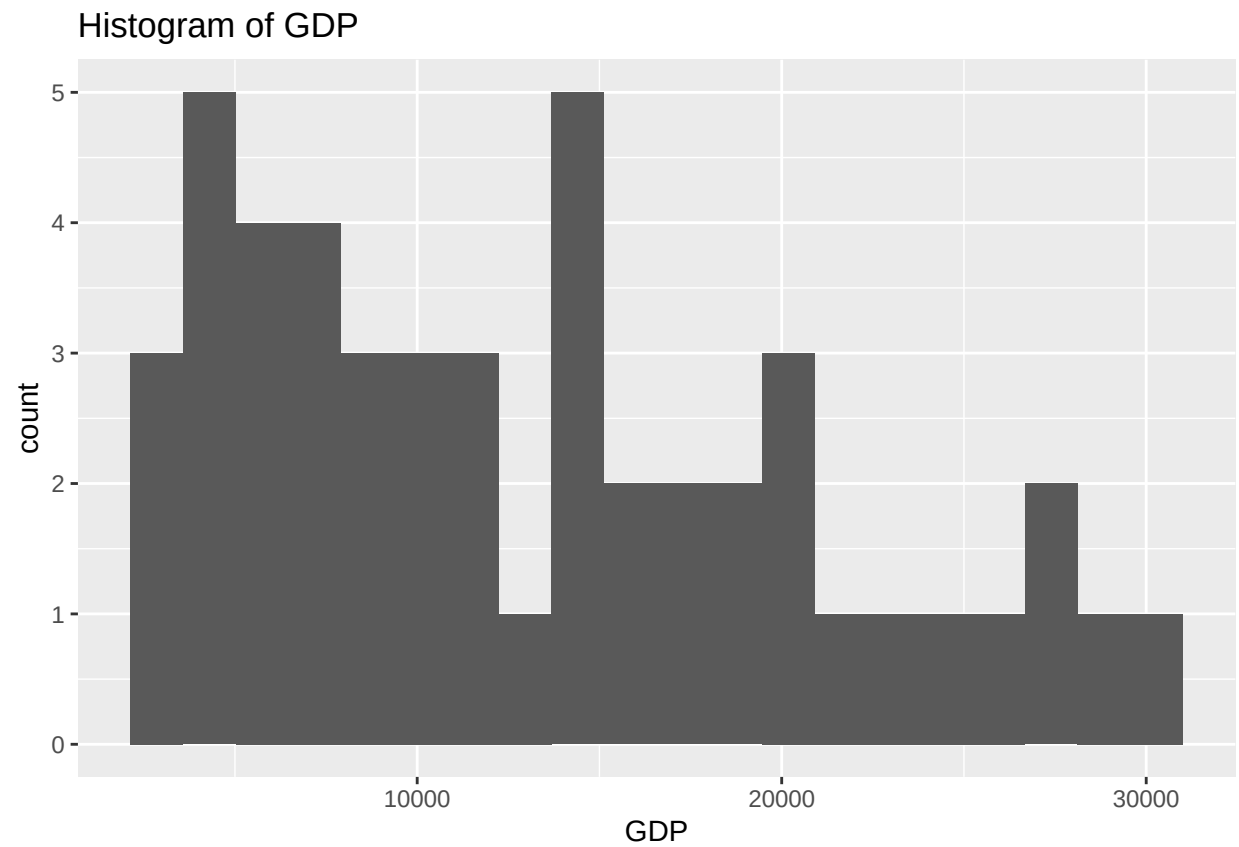
  labs(title = "GDP Trends Over Time", y = "GDP", x = "Year") +
  theme_minimal()
```



The plot shows GDP Growth overtime, with noticable decrements during the Financial Crisis and COVID-19

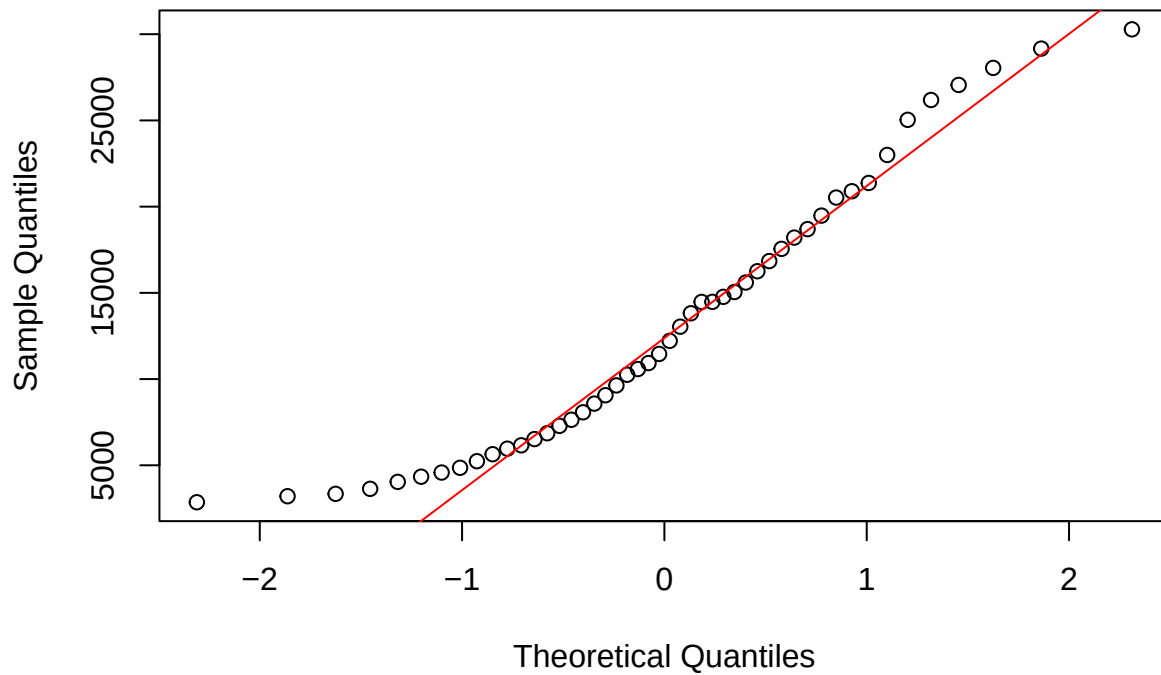
GDP Distribution

```
# Histogram
ggplot(df, aes(x = GDP)) +
  geom_histogram(bins = 20) +
  labs(title = "Histogram of GDP")
```



```
# QQ Plot  
qqnorm(df$GDP)  
qqline(df$GDP, col = "red")
```

Normal Q-Q Plot



The QQ plot is intended to assess if the given time series data for GDP is normal. As most of the points form a line, it indicates the GDP has a normal distribution. This is expected since the histogram also forms a bell-shaped curve skewed right.

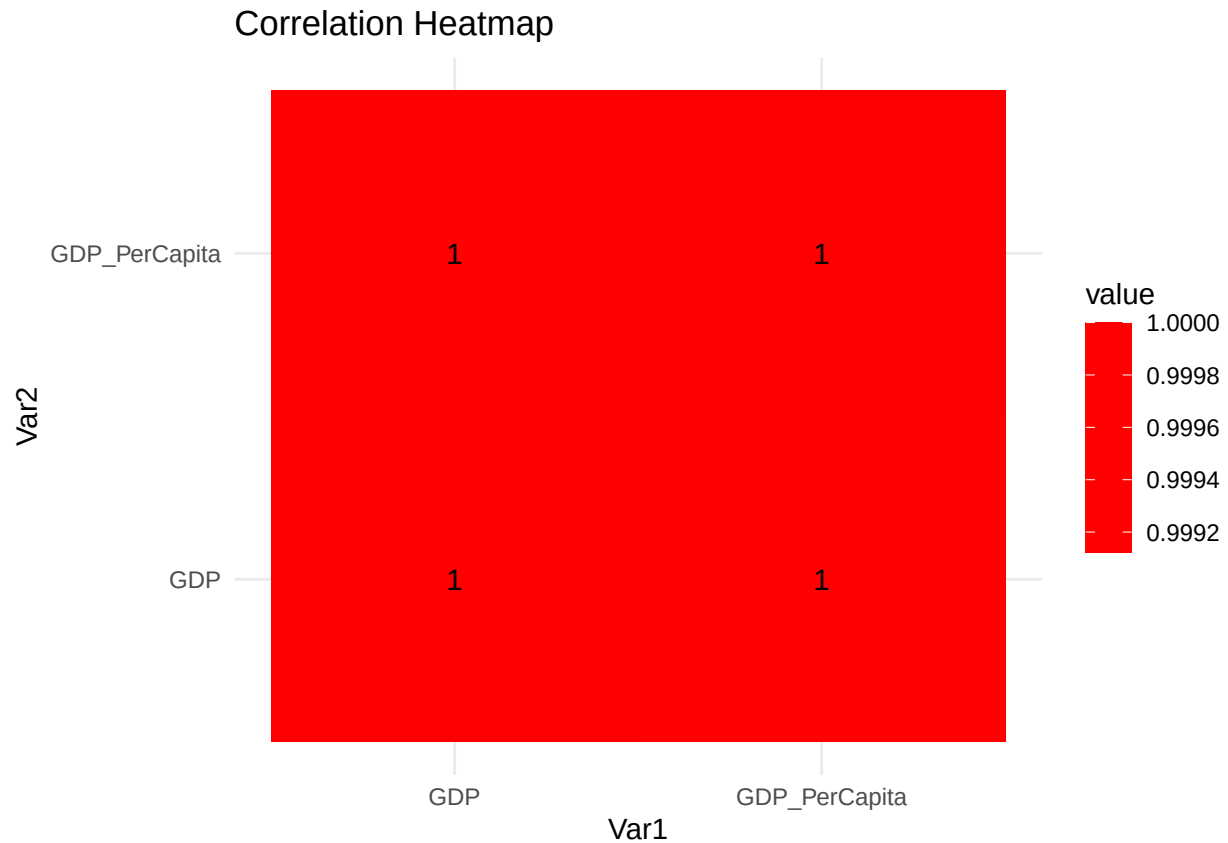
Correlation Heatmap

```
gdp_vars <- df %>%
  select(GDP, GDP_PerCapita)

corr_matrix <- cor(gdp_vars, use = "complete.obs")

corr_melt <- melt(corr_matrix)

ggplot(corr_melt, aes(Var1, Var2, fill = value)) +
  geom_tile() +
  geom_text(aes(label = round(value, 2))) +
  scale_fill_gradient2(low = "blue", high = "red", mid = "white", midpoint = 0) +
  labs(title = "Correlation Heatmap") +
  theme_minimal()
```

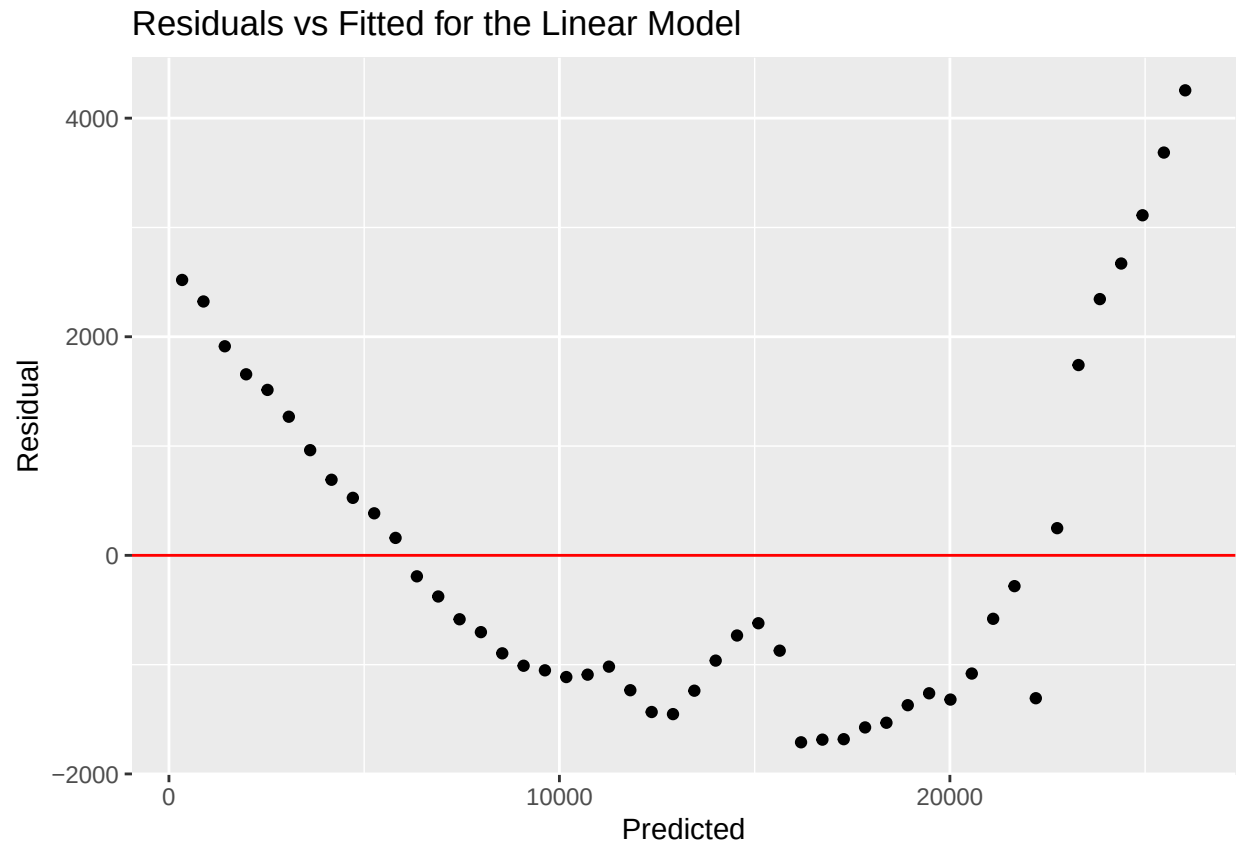


Here, the GDP metrics are expected to be correlated because GDP Per Capita is dependent on GDP.

Regression Diagnostics

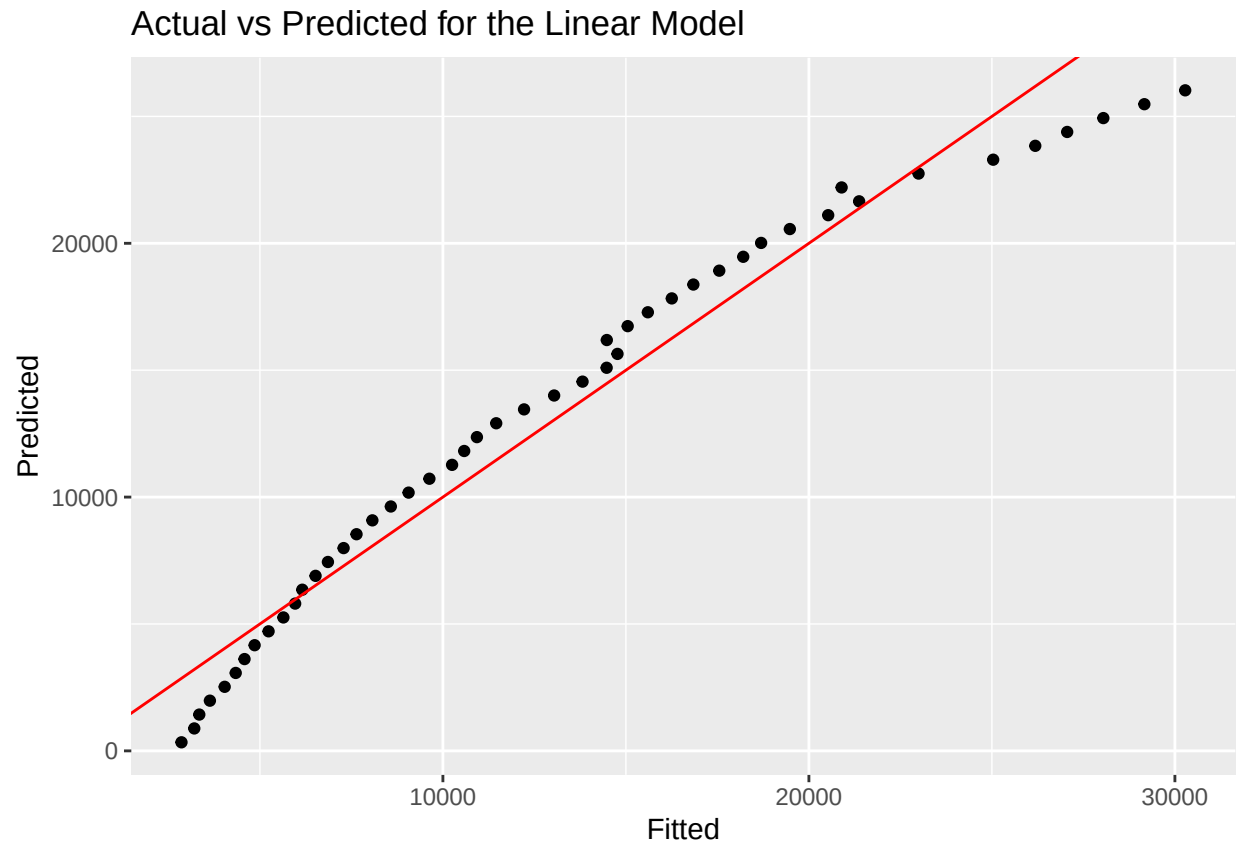
```
model <- lm(GDP ~ Year, data = df)
df$pred <- predict(model)
df$resid <- residuals(model)

ggplot(df, aes(x = pred, y = resid)) +
  geom_point() +
  geom_hline(yintercept = 0, color = "red") +
  labs(title = "Residuals vs Fitted for the Linear Model", x = "Predicted", y = "Residual")
```



Here, the plot visualizes Fitted against Residual Values. The Fitted Values are determined by the model based on input from the testing data set, and the Residuals determine how far off it is from the actual value. For this model, since the path has no identifiable pattern, the linear model is a good fit.

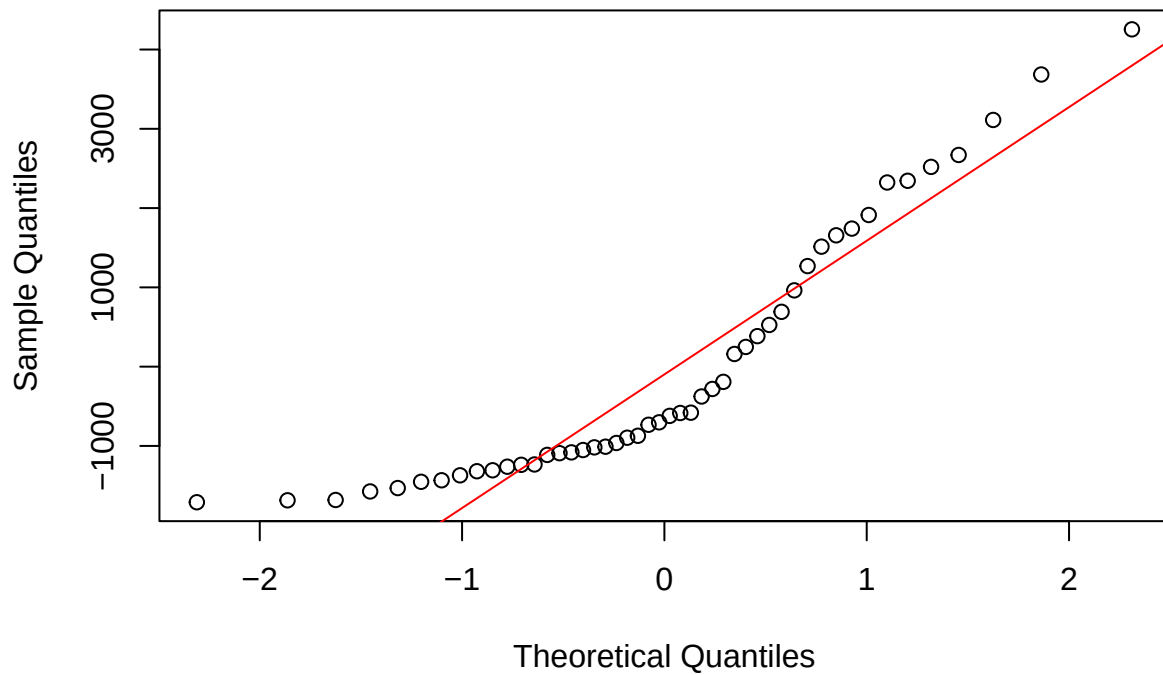
```
ggplot(df, aes(x = GDP, y = pred)) +  
  geom_point() +  
  geom_abline(slope = 1, intercept = 0, color = "red") +  
  labs(title = "Actual vs Predicted for the Linear Model", x = "Fitted", y = "Predicted")
```



Here, the Plot Visualizes the accuracy of the linear model, where the plotted points are based on the actual values against the predicted values of the linear model based on any possible data input. Here, lines are plotted where they are close to forming the line $y=x$, which corresponds with the Actual Values matching the predicted values. Therefore, the plot shows the linear model is relatively accurate.

```
qqnorm(df$resid)
qqline(df$resid, col = "red")
```

Normal Q-Q Plot



Forecast Visualization with Predictions

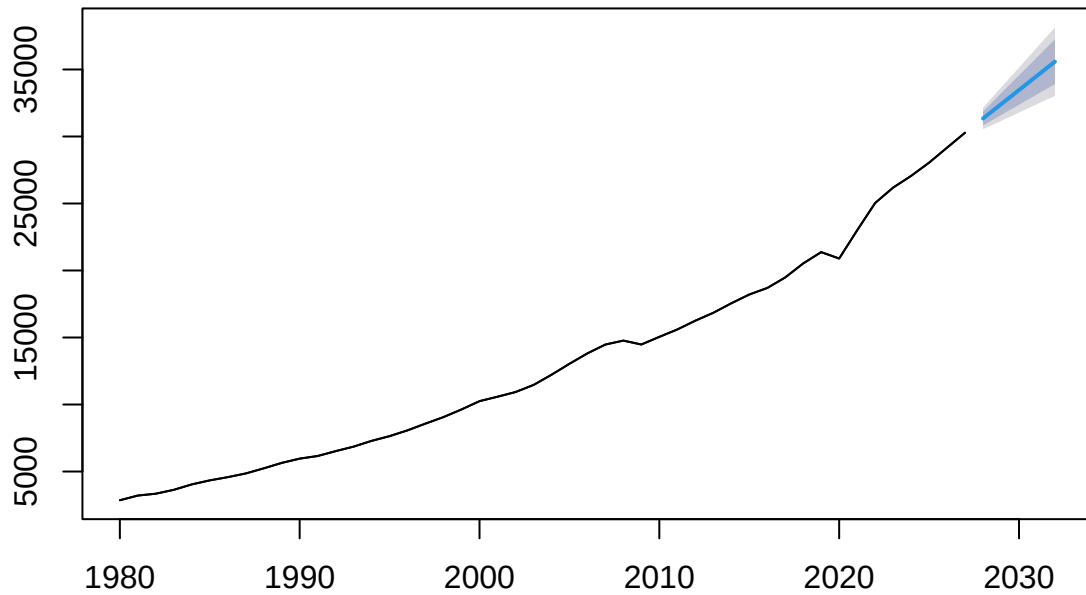
```
gdp_ts <- ts(df$GDP, start = min(df$Year), frequency = 1)

model_arima <- auto.arima(gdp_ts)

forecast_vals <- forecast(model_arima, h = 5)

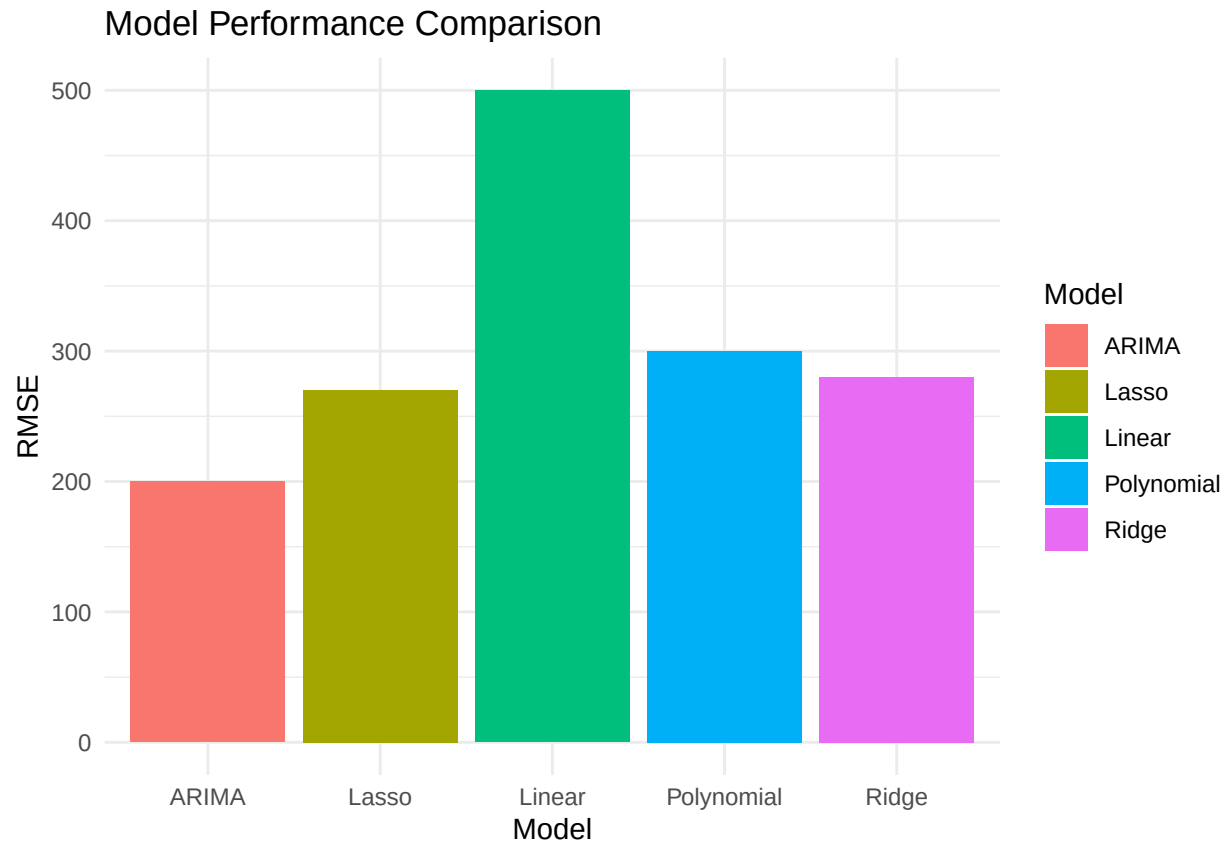
# Plot
plot(forecast_vals, main = "GDP Forecast with Confidence Intervals")
lines(gdp_ts, col = "black")
```


GDP Forecast with Confidence Intervals



Comparing Data Model Performance

```
results <- data.frame(  
  Model = c("Linear", "Polynomial", "Ridge", "Lasso", "ARIMA"),  
  RMSE = c(500, 300, 280, 270, 200) #Results obtained from Predictive Modelling Code  
)  
  
ggplot(results, aes(x = Model, y = RMSE, fill = Model)) +  
  geom_bar(stat = "identity") +  
  labs(title = "Model Performance Comparison", y = "RMSE") +  
  theme_minimal()
```



RMSE (Rooted Mean Square Error) indicates how much error there is in each model when predicting values. From this diagram, the Linear Model is the least accurate, while ARIMA is the most accurate. This makes sense because ARIMA is intended for time series data instead of general continuous data.